THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · HARD LEVEL FULL-YEAR PAPER SET 3

Full Syllabus (Chapters 1–10) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. A pattern adds the same number every time. Its first term is 7 and its 14th term is 150. The number added each time is: 1

(a) 10 **(b)** 11 **(c)** 12 **(d)** 13

2. The minute hand of a clock takes how many minutes to turn through 150°? 1

(a) 30 **(b)** 12 **(c)** 5 **(d)** 25

3. A 5-digit number ends with digit 7 and has digit sum 31; the sum of its other four digits is: 1

(a) 15 **(b)** 38 **(c)** 33 **(d)** 24

4. Out of 120 students, 45 play cricket and 38 play football, none plays both. How many play neither? 1

(a) 82 **(b)** 47 **(c)** 37 **(d)** 75

5. For the 3-digit number 1d3, the digit d that makes it divisible by 9 (smallest such d) is: 1

(a) 7 **(b)** 5 **(c)** 6 **(d)** 14

6. A wire in the shape of a square of side 20 cm is straightened and bent into a rectangle of length 26 cm. The breadth (cm) is: 1

(a) 14 **(b)** 8 **(c)** 26 **(d)** 6

7. If $\frac{3}{5}$ of a number is 12, then $\frac{1}{4}$ of the number is: 1

(a) 5 **(b)** 20 **(c)** 12 **(d)** 1

8. In rectangle ABCD, AC = BD = 8 cm. If O is the intersection, the sum OA + OB + OC + OD (cm) is: 1

(a) 20 **(b)** 12 **(c)** 24 **(d)** 16

9. A figure has rotational symmetry of order 3 and one of its lines of symmetry. The total number of lines of symmetry it must have is: 1

(a) 4 **(b)** 3 **(c)** 2 **(d)** 5

10. A trader loses ₹150 on day 1, gains ₹240 on day 2 and loses ₹120 on day 3. The net result in ₹ is: 1

(a) -32 **(b)** -30 **(c)** -28 **(d)** -20

11. A pattern adds 1, then 2, then 3, ... Starting at 10, the term after 12 additions is: 1

(a) 154 **(b)** 88 **(c)** 76 **(d)** 22

12. 4 lines are drawn through a point O. The number of straight angles at O (pairs of opposite rays) is: 1

(a) 5 **(b)** 4 **(c)** 12 **(d)** 3

13. The number of digits in the largest palindrome smaller than 10000 is: 1

(a) 3 **(b)** 4 **(c)** 6 **(d)** 5

14. A bar graph shows sales of ₹2000, ₹3500, ₹2500. If the scale is 1 cm = ₹500, the total height of all three bars is (cm): 1

(a) 18 **(b)** 14 **(c)** 8000 **(d)** 16

15. Which is the greatest 3-digit number that leaves remainder 3 when divided by 7? 1

(a) 990 **(b)** 1004 **(c)** 998 **(d)** 997

16. Tiling a rectangular floor of length 9 m at ₹150 per m^2 costs ₹6750. Its breadth (m) is: 1

(a) 10 **(b)** 45 **(c)** 14 **(d)** 5

17. **Assertion (A):** A square has 4 lines of symmetry. 1
Reason (R): A regular polygon with n sides has n lines of symmetry.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

18. **Assertion (A):** $-7 < -3$. 1
Reason (R): On a number line, a number lying to the left of another number is smaller.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

19. **Assertion (A):** The sum of the first 5 odd numbers is 25. 1
Reason (R): 25 is a square number.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

20. **Assertion (A):** A straight angle measures $180°$. 1
Reason (R): A right angle measures $90°$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. Find the number of terms in the hailstone (Collatz) sequence of 11 and its largest term. 2

22. A bar graph uses 1 cm = 20 units and a bar is 9 cm. Another bar is 3 cm shorter. Find the value of the shorter bar. 2

23. Show that the product of 3 consecutive numbers starting from 14 is divisible by 3. 2

24. A rectangular field is 40 m by 30 m. Find the cost of levelling the field at ₹5 per m^2 and fencing at ₹12 per metre. 2

25. Find the fraction which must be added to $\frac{2}{7}$ to get $\frac{5}{6}$. 2

26. A rectangle has diagonal intersection O with $\angle AOB = 35^\circ$. Find all four angles at O. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. Find the missing numerator ?: $\frac{?}{12} + \frac{1}{4} = \frac{5}{6}$. 3

28. Two concentric circles have radii 4 cm and 9 cm. Find the width of the ring, the diameter of the inner circle and the length of the longest chord of the outer circle. 3

29. A regular polygon has 4 sides (4 is even). Of its 4 lines of symmetry, how many pass through two opposite vertices and how many pass through the midpoints of two opposite sides? 3

30. The sum of two integers is -4. One integer is 11. Find the other and their difference (larger minus smaller). 3

31. A tower is built in rows of cubes: row 1 has 1 cube, row 2 has 3, row 3 has 6, and so on (triangular numbers). How many cubes are in row 6, and how many in the first 4 rows together? 3

OR

A pizza is cut by 2 lines through the centre into equal slices (all lines passing through the centre). Find the angle of each slice and the number of slices. 3

32. How many 4-digit numbers can be formed from 2, 5, 7, 9 using each digit once? What is the largest and the smallest? What is their difference? 3

OR

For the data 5, 5, 2, 7, 8, 3, 5, 1, 7, 4, 6, 5 prepare a frequency table and find how many values are greater than 5. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. **Case study.** Two temple bells ring together at 9:00 am. One bell rings every 12 minutes and the other every 18 minutes. 5

(i) Write 12 and 18 as products of prime factors.
(ii) Find the LCM of 12 and 18.
(iii) At what time will they ring together next?
(iv) How many more times will they ring together before 11:00 am? Give the times.

34. **Case study.** Farmer Ramesh has a rectangular field 40 m long and 25 m wide. He will fence it with wire and then sow seeds. 5

(i) Find the perimeter of the field.
(ii) Find the area of the field.
(iii) Find the cost of fencing at ₹20 per metre.
(iv) 2 kg of seeds are needed for every 50 m^2. How many kg of seeds does he need?

35. In a pattern of dot figures, figure 1 has 5 dots and each next figure has 9 more dots than the one before. How many dots are in figure 15? Which figure has 176 dots? 5

36. At 4 o'clock, find 5

(a) the angle turned by the hour hand since 12 o'clock,

(b) the angle turned by the minute hand since 12 o'clock,

(c) the smaller angle between the hands,

(d) its type.

OR

A palindromic number of 5 digits is formed as $abcba$ where $a = 2$, $b = 4$, $c = 5$. Write the number, find its digit sum, check if it is divisible by 9, and find the smallest number to add to make it divisible by 9. 5

37. A bar graph has scale 1 cm = 15 units. Bars A, B, C have heights 6 cm, 4 cm, 8 cm. The value of D equals A + B - C. Find the values of A, B, C and D, and the height of bar D. 5

OR

A rectangular garden is 72 m by 48 m. Square flower beds of the largest possible equal size are to be made covering the garden. Find 5

(i) the side of each bed,

(ii) the number of beds,

(iii) the cost of fencing all beds at ₹4 per metre if each bed is fenced separately.

38. A rectangle with length 24 cm and breadth 10 cm has a triangular piece with base 24 cm and height 6 cm cut from one side. Find the area of the remaining piece and the fraction of the whole that is left. 5

OR

A trader sold $\frac{1}{4}$ of his cloth on day 1 and $\frac{2}{3}$ of the remaining on day 2. If 12 metres are left, find the total length of cloth. 5

END OF QUESTION PAPER **Total: 80 Marks**

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